IS601 | MODULE 10

User Models and Deployment

SQLAlchemy, PostgreSQL, and Docker Hub

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Source commit: d6eef8339601790f4ab57d5e2d2cbba67b20c783

GitHub repository

https://github.com/tkg1121/fastapi-postgres-calculator

GitHub Actions workflow

https://github.com/tkg1121/fastapi-postgres-calculator/actions/runs/42955563012

Docker Hub image

https://hub.docker.com/r/tkg1121/fastapi-postgres-calculator

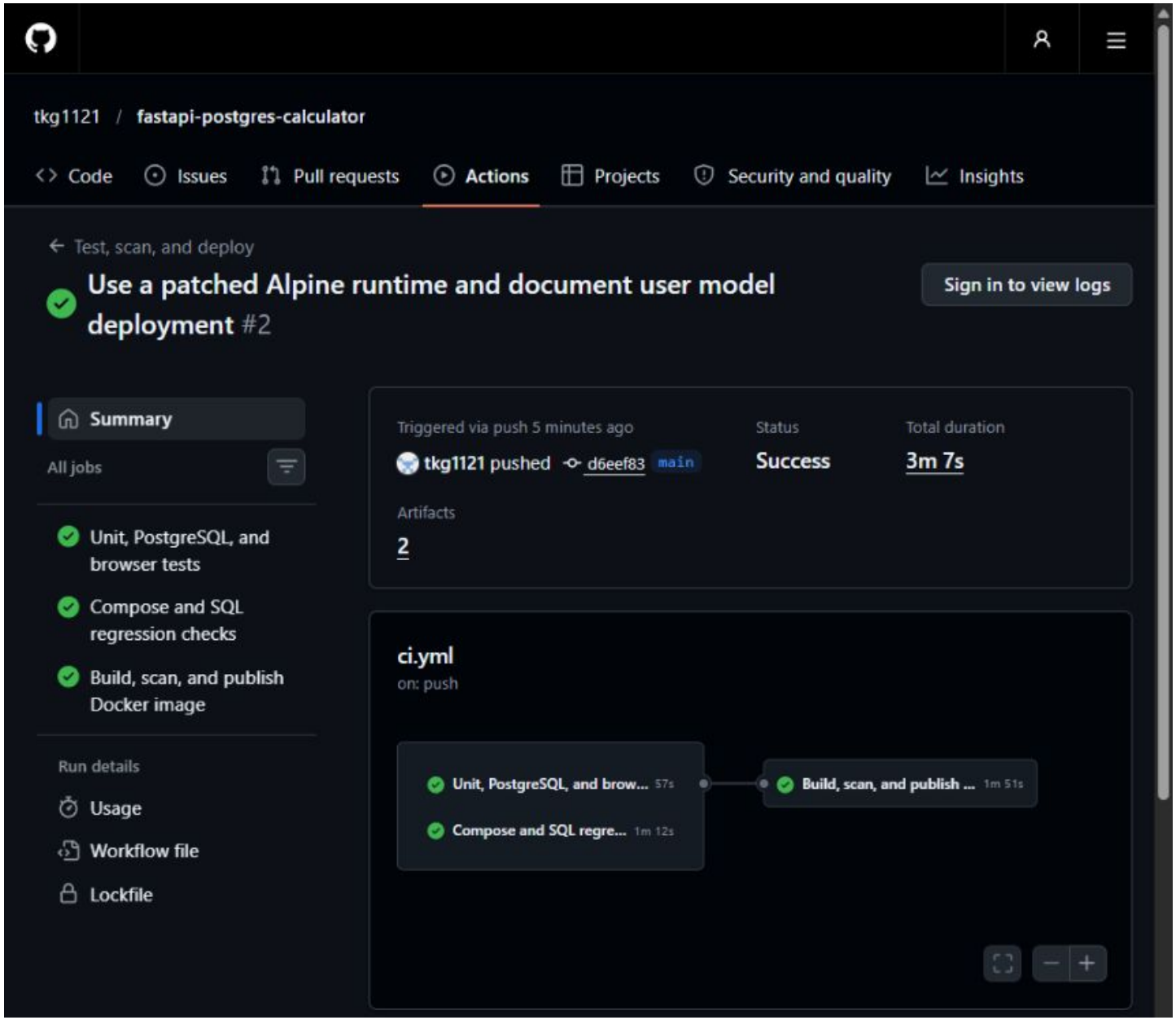
This assignment extends the existing calculator with a SQLAlchemy User model, Pydantic request and response schemas, and salted Argon2id password hashing. User responses include id, username, email, and created_at. Passwords and password hashes are excluded.

Alembic creates calculator.users without changing the earlier public.users and public.calculations tables. The API supports registration and lookup, including validation, duplicate-record handling, and database error responses.

The suite contains 206 tests covering unit, API/PostgreSQL integration, and browser testing. The CI workflow also runs the complete Compose stack, scans the built image for HIGH and CRITICAL vulnerabilities, and checks the published image after pulling it from Docker Hub.

GitHub Actions workflow

The workflow runs unit, PostgreSQL, browser, and Compose checks before building, scanning, and publishing the image. The published image is pulled and checked through its API.



Docker Hub deployment

Docker Hub shows the published latest tag, its image digest, the linux/amd64 platform, and a pull command. A separate commit-specific tag identifies the source version.

